

Requirements

Functional and non-functional requirements

Research Assistant - local-first multi-agent research workspace
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Research Assistant - Functional & Non-Functional Requirements

Project: Local-first Multi-Agent Research Assistant **Version:** 1.0 **Status:** Draft for implementation

1. Product Summary

A privacy-preserving, local-first multi-agent system that helps a researcher discover, ingest, analyse, compare and interrogate academic papers. All model inference runs on a self-hosted Ollama server so that research content never leaves the user's infrastructure. Only public scholarly metadata APIs are contacted over the network.

1.1 Core Concepts

Concept	Definition
Research Space	A named, long-lived container holding one conversation thread, its pinned papers, notes and derived artefacts. Equivalent to a "project".
Paper	A bibliographic record (metadata + optional full text + parsed chunks + embeddings).
Pin	The act of attaching a Paper to a Research Space, making it part of that space's retrieval corpus.
Note	User-authored or agent-generated structured annotation linked to a Paper and a Space.
Turn	One user message + the system's grounded response, persisted in the Space.

2. Functional Requirements

Priority key: **M** = Must (MVP), **S** = Should, **C** = Could, **W** = Won't (this release). Phase key indicates target delivery phase (P1-P4).

FR-1 Discovery & Search

ID	Requirement	Pri	Phase
FR-1.1	The system shall accept a free-text research topic and return relevant academic papers.	M	P1

ID	Requirement	Pri	Phase
FR-1.2	The system shall query multiple scholarly sources in parallel: arXiv, Semantic Scholar, OpenAlex, CORE, PubMed/PMC, CrossRef.	M	P1/P2
FR-1.3	The system shall deduplicate results across sources using DOI, arXiv ID, and normalised title+author fuzzy match.	M	P1
FR-1.4	The system shall support exact-paper lookup when the user supplies a paper title, DOI, arXiv ID, or PMID.	M	P2
FR-1.5	The system shall rank results semantically against the user's intent, not only by keyword match, recency or citation count.	M	P1
FR-1.6	The ranking shall combine: dense semantic similarity, cross-encoder rerank score, citation count, recency, and open-access availability, using a configurable weighted formula.	S	P1
FR-1.7	The system shall expose filters: year range, source, open-access only, venue, author, minimum citations.	S	P2
FR-1.8	The system shall display for each result: title, authors, year, venue, abstract, citation count, source, OA status, and a direct link.	M	P1
FR-1.9	The system shall support general web/knowledge-base search as a supplementary source for context that is not in paper indices.	C	P3
FR-1.10	The system shall cache external API responses per normalised query to reduce latency and respect rate limits.	M	P1

ID	Requirement	Pri	Phase
FR-1.11	The system shall degrade gracefully when one or more external sources are unavailable, returning partial results with a source-health indicator.	M	P1

FR-2 Ingestion & Library

ID	Requirement	Pri	Phase
FR-2.1	The user shall be able to pin/attach any discovered paper to the active Research Space.	M	P1
FR-2.2	The user shall be able to upload a local PDF and have it treated as a first-class Paper.	M	P1
FR-2.3	The system shall retrieve open-access full text where legally available; otherwise it shall fall back to abstract + metadata and clearly label reduced coverage.	M	P1
FR-2.4	The system shall parse PDFs with layout awareness, preserving section headings, page numbers, tables and reference lists.	M	P1
FR-2.5	The system shall OCR scanned/image-only PDFs as a fallback.	S	P2
FR-2.6	The system shall chunk documents semantically (section/paragraph aware) with configurable size and overlap.	M	P1
FR-2.7	The system shall generate embeddings via the self-hosted Ollama embedding model and store them in a space-scoped vector collection.	M	P1

ID	Requirement	Pri	Phase
FR-2.8	Ingestion shall run asynchronously with visible progress states: queued -> fetching -> parsing -> chunking -> embedding -> ready -> failed.	M	P1
FR-2.9	The system shall deduplicate ingestion: a paper already embedded shall be reused across Spaces without re-embedding.	S	P2
FR-2.10	The user shall be able to unpin or delete a paper, which removes it from retrieval for that Space.	M	P1

FR-3 Conversational Q&A (RAG)

ID	Requirement	Pri	Phase
FR-3.1	The user shall be able to ask natural-language questions answered strictly from the papers pinned to the active Space.	M	P1
FR-3.2	Every factual claim in an answer shall carry an inline citation resolving to paper + section + page + chunk.	M	P1
FR-3.3	Retrieval shall be hybrid: dense vector search plus BM25 sparse search, fused (RRF) and then cross-encoder reranked.	M	P3
FR-3.4	The system shall explicitly answer "not supported by the attached papers" rather than speculate when evidence is insufficient.	M	P1
FR-3.5	Answers shall be produced through the Phase 3 peer-review verification loop (see §4).	M	P3

ID	Requirement	Pri	Phase
FR-3.6	The system shall surface a confidence/groundedness score and the number of verification iterations performed.	S	P3
FR-3.7	The user shall be able to click any citation and view the exact source passage highlighted in context.	S	P2
FR-3.8	The system shall stream answer tokens to the UI as they are produced.	M	P1
FR-3.9	The user shall be able to scope a question to a subset of pinned papers.	S	P2

FR-4 Comparison & Gap Analysis

ID	Requirement	Pri	Phase
FR-4.1	The system shall extract a structured profile per paper: problem, objective, method, dataset, metrics, results, limitations, future work.	M	P2
FR-4.2	The system shall generate a comparison matrix across any selected set of pinned papers.	M	P2
FR-4.3	The system shall identify commonalities (shared methods, datasets, findings, assumptions).	M	P2
FR-4.4	The system shall identify contradictions and disagreements between papers, with citations for both sides.	M	P2
FR-4.5	The system shall identify research gaps: unexplored combinations, acknowledged limitations not addressed elsewhere, missing baselines, under-studied populations/datasets.	M	P2

ID	Requirement	Pri	Phase
FR-4.6	Comparison output shall be exportable (Markdown, CSV, PDF).	S	P3
FR-4.7	Comparison results shall be cached and invalidated when the pinned set changes.	S	P2
FR-4.8	Gap analysis output shall also pass the peer-review verification loop.	S	P3

FR-5 Notes & Annotation

ID	Requirement	Pri	Phase
FR-5.1	The system shall auto-generate structured notes per paper (summary, key contributions, methodology, results, limitations, relevance to the Space's topic).	M	P2
FR-5.2	The user shall be able to create, edit, and delete free-form notes attached to a Paper or a Space.	M	P2
FR-5.3	Notes shall be searchable and shall be retrievable as context in Q&A.	S	P2
FR-5.4	The user shall be able to highlight a passage and attach a note to it.	C	P3
FR-5.5	Notes shall be exportable per paper or per Space.	S	P3

FR-6 Memory, Context & Sessions

ID	Requirement	Pri	Phase
FR-6.1	Conversations shall persist indefinitely and be resumable across application restarts.	M	P1
FR-6.2	Papers discovered or pinned in a Space shall remain bound to that Space.	M	P1

ID	Requirement	Pri	Phase
FR-6.3	The system shall maintain long-conversation context via rolling summarisation plus vector recall of earlier turns, without unbounded context growth.	M	P2
FR-6.4	The system shall maintain a per-Space "findings memory": accumulated conclusions, open questions and decisions.	S	P3
FR-6.5	The user shall be able to rename, archive, duplicate and delete a Research Space.	S	P2
FR-6.6	Full-text search across all conversations and notes shall be available.	C	P3

FR-7 Administration & Model Management

ID	Requirement	Pri	Phase
FR-7.1	The system shall connect to a configurable self-hosted Ollama endpoint.	M	P1
FR-7.2	Chat model, embedding model, reranker model and verifier model shall be independently configurable.	M	P1
FR-7.3	The system shall expose health checks for Ollama, vector store, database, queue and each external API.	M	P1
FR-7.4	The system shall support multiple users with strict data isolation.	S	P4
FR-7.5	An evaluation dashboard shall report retrieval quality, groundedness, verification loop statistics and latency percentiles.	S	P4

3. Non-Functional Requirements

NFR-1 Performance & Latency

ID	Requirement	Target
NFR-1.1	Time to first streamed token for a Q&A turn	≤ 2.5 s (p50), ≤ 5 s (p95)
NFR-1.2	Complete Q&A turn including verification loop	≤ 12 s (p50), ≤ 25 s (p95)
NFR-1.3	Discovery search results returned	≤ 3 s (p50), ≤ 8 s (p95)
NFR-1.4	Retrieval stage (hybrid search + rerank)	≤ 800 ms (p95)
NFR-1.5	Ingestion of a 20-page PDF end-to-end	≤ 60 s (p95), asynchronous
NFR-1.6	Comparison of 10 pinned papers	≤ 90 s (p95), asynchronous with progress
NFR-1.7	UI interaction responsiveness (non-inference)	≤ 200 ms

Design note: external API fan-out is parallel; embeddings are computed once and cached; verification loop is bounded to a maximum iteration count so worst-case latency is deterministic.

NFR-2 Accuracy & Quality

ID	Requirement	Target
NFR-2.1	Groundedness - every claim traceable to a retrieved passage	$\geq 95\%$ of claims on attached-paper Q&A
NFR-2.2	Answer correctness on curated evaluation set	$\geq 90\%$
NFR-2.3	Citation precision - cited passage genuinely supports the claim	$\geq 95\%$
NFR-2.4	Retrieval recall@10 on evaluation set	$\geq 90\%$
NFR-2.5	Abstention correctness - system declines when evidence absent	$\geq 95\%$ of unanswerable probes
NFR-2.6	Hallucinated-citation rate (citation to non-existent/irrelevant source)	$\leq 1\%$
NFR-2.7	Semantic ranking quality, nDCG@10 on labelled topic set	≥ 0.80

Design note: these are engineered targets enforced by the Phase 3 verification loop and measured continuously by the evaluation harness; they are not model guarantees.

NFR-3 Scalability

ID	Requirement
NFR-3.1	API, orchestrator and worker tiers shall be stateless and horizontally scalable.
NFR-3.2	Ollama inference shall run as an independently scalable GPU pool behind a load balancer.
NFR-3.3	The vector store shall support $\geq 10M$ chunks with sub-second filtered search.
NFR-3.4	The system shall support ≥ 100 concurrent active Research Spaces per node group.
NFR-3.5	Ingestion workers shall scale on queue depth without affecting chat latency.
NFR-3.6	Per-Space vector isolation shall be achieved via collection/namespace partitioning.

NFR-4 Privacy & Security

ID	Requirement
NFR-4.1	No paper content, note, or conversation shall be transmitted to any third-party LLM provider.
NFR-4.2	Outbound network access shall be limited to an allow-list of scholarly metadata/full-text endpoints.
NFR-4.3	All inference shall occur on the self-hosted Ollama deployment.
NFR-4.4	Data at rest (database, vector store, uploaded PDFs) shall be encryptable.
NFR-4.5	Authentication shall use JWT/OIDC; all API routes except health shall be authenticated.
NFR-4.6	Strict tenant isolation shall be enforced at query level for every data access.
NFR-4.7	User inputs and retrieved document text shall be sanitised to mitigate prompt injection from malicious PDFs.
NFR-4.8	Agent tool invocations shall be validated against an allow-listed schema.

NFR-5 Reliability & Availability

ID	Requirement
NFR-5.1	Target availability 99.5% for the API tier.

ID	Requirement
NFR-5.2	External source failures shall be isolated by circuit breakers and shall not fail the request.
NFR-5.3	Ingestion jobs shall be idempotent and retried with exponential backoff (max 3 attempts).
NFR-5.4	The verification loop shall be bounded and shall always return a best-effort, clearly-labelled answer.
NFR-5.5	Database and vector store shall be backed up daily with point-in-time recovery.

NFR-6 Observability

ID	Requirement
NFR-6.1	Every agent hop shall emit a trace span (input, output, model, tokens, latency).
NFR-6.2	Verification loop iterations, reviewer verdicts and rejection reasons shall be logged per turn.
NFR-6.3	Retrieval diagnostics (queries, candidates, scores, selected chunks) shall be inspectable per answer.
NFR-6.4	Latency, token throughput, GPU utilisation and queue depth shall be exported as metrics.
NFR-6.5	Structured JSON logs with correlation IDs across API -> orchestrator -> agent -> model.

NFR-7 Maintainability & Portability

ID	Requirement
NFR-7.1	Agents shall be independently testable units with typed input/output contracts.
NFR-7.2	Model, vector store and paper-source integrations shall sit behind provider interfaces to avoid lock-in.
NFR-7.3	Prompts shall be versioned, externalised from code, and diffable.
NFR-7.4	The full stack shall run via a single <code>docker compose up</code> for local development.
NFR-7.5	Minimum 80% unit-test coverage on agent and retrieval logic.

NFR-8 Compliance & Ethics

ID	Requirement
NFR-8.1	Only open-access or user-supplied full text shall be stored; paywalled content shall not be scraped.
NFR-8.2	External API terms of use and rate limits shall be respected, with a polite User-Agent and contact email.
NFR-8.3	Generated summaries shall always attribute the source paper.
NFR-8.4	The UI shall state that outputs are AI-generated and require scholarly verification.

NFR-9 Usability & Accessibility

ID	Requirement
NFR-9.1	Citations shall be one click from claim to source passage.
NFR-9.2	Long-running operations shall always show determinate progress.
NFR-9.3	WCAG 2.1 AA conformance.
NFR-9.4	Keyboard-first navigation for search, pin and chat.

4. Phase 3 - Loop-Engineered Accuracy (Peer-Review Verification)

4.1 Objective

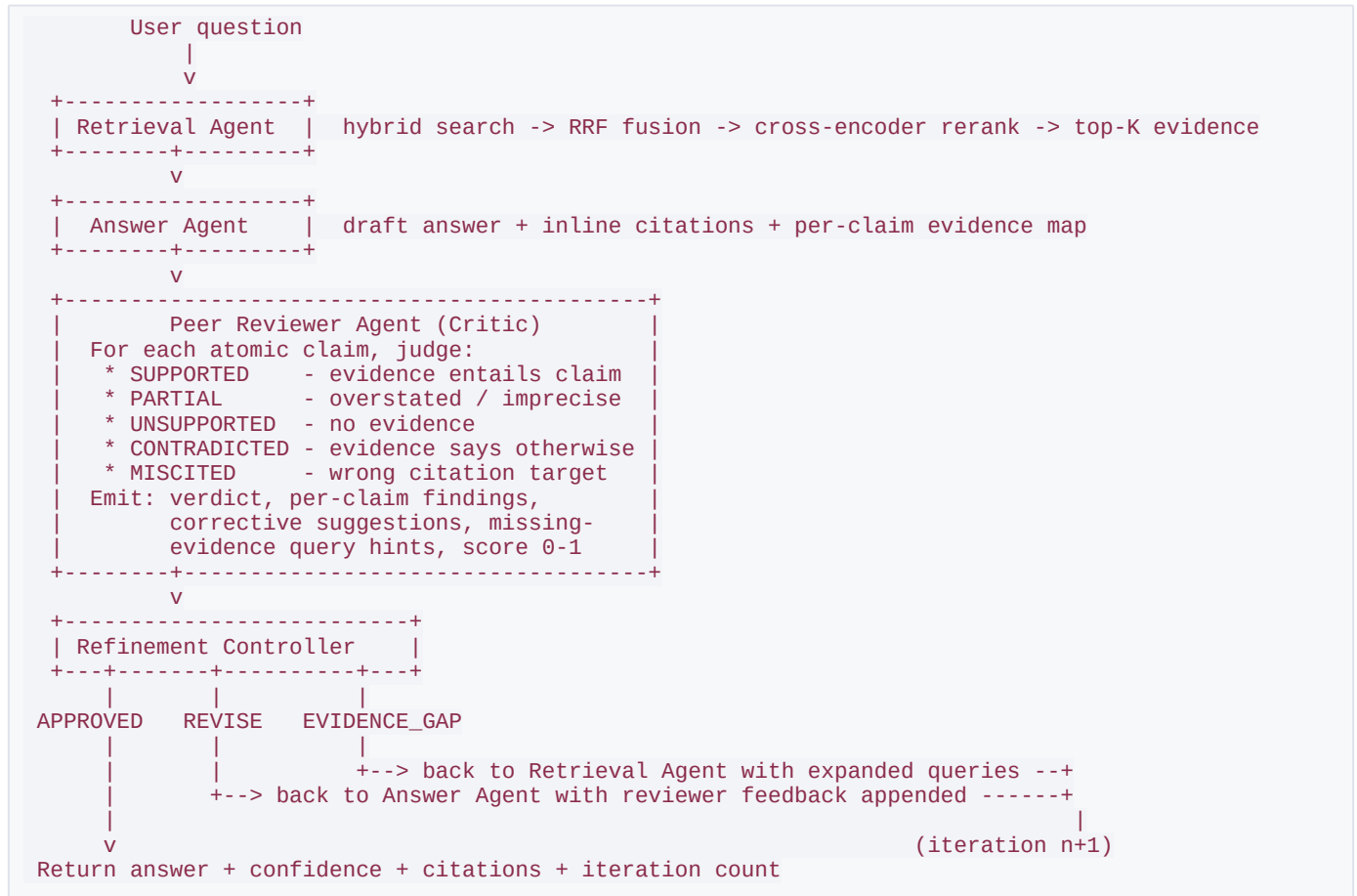
No answer reaches the user until a **peer reviewer agent** has independently verified it against the retrieved evidence. If the reviewer rejects it, the answer is regenerated **with the reviewer's specific corrective feedback** and re-verified, repeating until the reviewer approves or the iteration budget is exhausted.

4.2 Loop Participants

Agent	Responsibility
Answer Agent (Generator)	Produces a cited, evidence-grounded draft answer from retrieved context.
Peer Reviewer Agent (Critic)	Independently verifies the draft claim-by-claim against the same evidence. Runs on a separate prompt and may run on a different/larger model. Never sees the generator's reasoning - only the question, evidence and draft, to avoid confirmation bias.
Refinement Controller	Interprets reviewer verdicts, decides next action (accept / regenerate / re-retrieve / abstain), enforces the iteration budget and convergence rules.

Agent	Responsibility
Retrieval Agent	Re-invoked with reviewer-suggested query expansions when the failure is "evidence missing" rather than "reasoning wrong".

4.3 Loop Flow



4.4 Loop Control Rules

Rule	Specification
LC-1 Iteration budget	Maximum 3 refinement iterations (configurable 1-5).
LC-2 Acceptance threshold	Reviewer score ≥ 0.90 and zero UNSUPPORTED/CONTRADICTED/MISCITED claims.
LC-3 Routing	UNSUPPORTED/insufficient-evidence -> re-retrieve. PARTIAL/MISCITED/overstatement -> regenerate only. CONTRADICTED -> regenerate with explicit contradiction notice.
LC-4 Feedback carry-forward	Each iteration receives the full history of prior reviewer findings to prevent regression on already-fixed claims.
LC-5 Convergence guard	If reviewer score fails to improve by ≥ 0.05 across two consecutive iterations, stop early (oscillation detection).

Rule	Specification
LC-6 Exhaustion behaviour	On budget exhaustion, return the highest-scoring draft, strip or flag unverified claims, and surface a clear low-confidence warning. Never silently return a rejected answer.
LC-7 Abstention	If after re-retrieval no supporting evidence exists, return an explicit "not answerable from the attached papers" with what was found.
LC-8 Independence	Reviewer prompt, temperature and (optionally) model must differ from the generator's to avoid correlated errors. Reviewer temperature ~ 0.
LC-9 Determinism	Generator temperature <= 0.3 for factual Q&A; reviewer 0.0.
LC-10 Auditability	Every iteration (draft, verdict, findings, score, action) persisted and inspectable in the UI "Verification trail".
LC-11 Latency guard	Total loop wall-clock capped (default 20 s); on breach return best draft with a warning.
LC-12 Escalation	On iteration 3, the reviewer may be escalated to the larger model for a final adjudication pass.

4.5 Reviewer Output Contract

```
{
  "verdict": "APPROVED | REVISE | EVIDENCE_GAP | REJECT",
  "overall_score": 0.0,
  "claims": [
    {
      "claim_id": "c1",
      "claim_text": "...",
      "status": "SUPPORTED | PARTIAL | UNSUPPORTED | CONTRADICTED | MISCITED",
      "cited_chunk_ids": [...],
      "evidence_quote": "...",
      "issue": "...",
      "suggested_correction": "..."
    }
  ],
  "missing_evidence_queries": [...],
  "global_feedback": "...",
  "hallucination_flags": [...]"
}
```

4.6 Applicability

The loop is mandatory for: Paper Q&A answers (FR-3.5), gap-analysis conclusions (FR-4.8), and auto-generated paper notes. It is optional for: discovery result listing and UI-level summarisation of already-verified content.

4.7 Loop Quality Metrics

Metric	Purpose	Target
First-pass approval rate	Generator quality without help	>= 65%
Post-loop approval rate	End-to-end accuracy gate	>= 95%

Metric	Purpose	Target
Mean iterations per turn	Latency/cost control	≤ 1.6
Budget-exhaustion rate	Frequency of low-confidence returns	$\leq 5\%$
Reviewer false-approval rate (human-audited sample)	Reviewer reliability	$\leq 3\%$
Added latency from loop	Cost of accuracy	≤ 6 s (p50)

5. Acceptance Criteria (Release Gates)

1. All **M**-priority functional requirements for the phase are implemented and demonstrable.
2. Evaluation harness reports NFR-2 targets met on the curated gold-standard question set.
3. p95 latency budgets in NFR-1 met under 20 concurrent users.
4. Zero outbound transmission of paper/conversation content verified by network egress audit.
5. Verification trail visible and complete for every Phase 3 answer.
6. Loop metrics in §4.7 met over a minimum 500-turn evaluation run.